Assignment 3

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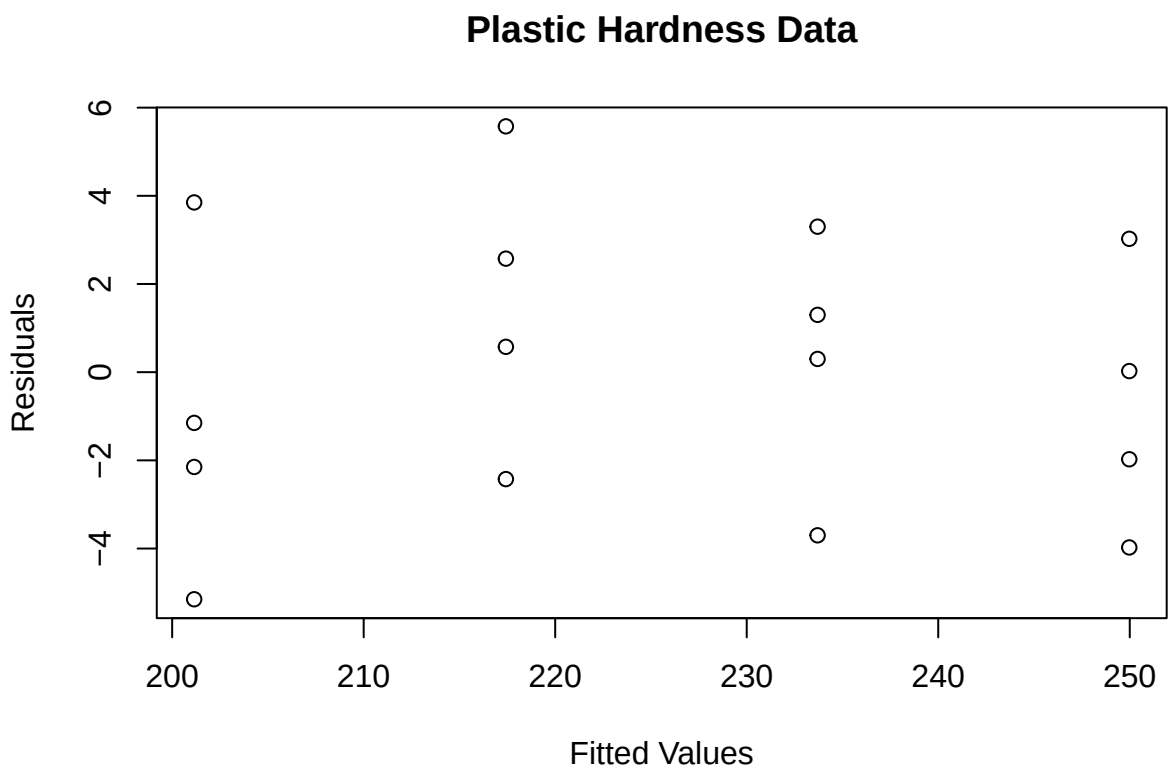
Problem 1

a-)

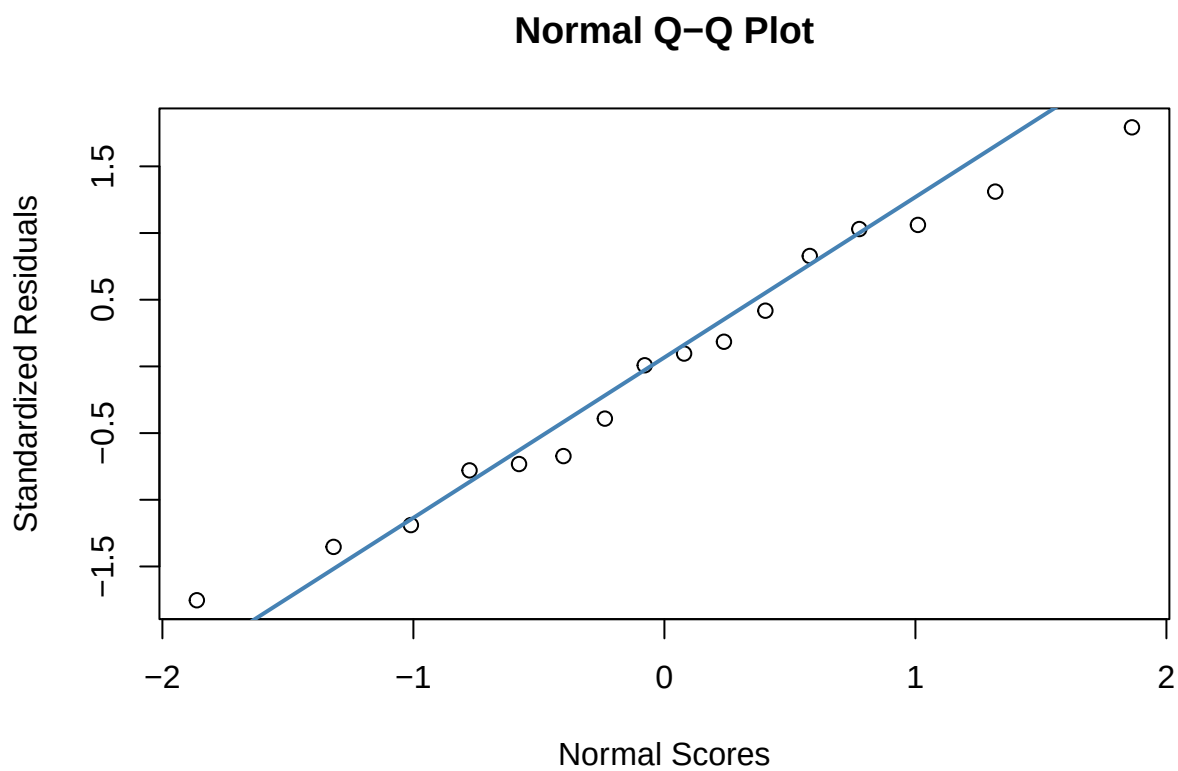
```
PH.data <- read.csv("/cloud/project/Data/Plastic Hardness Data.csv")
PH.reg <- lm(Y ~ X, data=PH.data)
ei <- PH.reg$residuals
plot(PH.reg$fitted.values, ei, xlab="Fitted Values", ylab="Residuals", main="Plastic Hardness Data")
```



The plot above does not indicate unequal variances, so no departures from regression model are evident here.

b-)

```
PHerror.std <- rstandard(PH.reg)
PHqq <- qqnorm(PHerror.std, xlab="Normal Scores", ylab="Standardized Residuals")
qqline(PHerror.std, col="steelblue", lwd=2)
```




```
cc <- cor(PHqq$x, PHqq$y) #0.991
```

The normality assumption appears to be reasonable as the above normal probability plot indicates. The coefficient of correlation between residuals and their expected values under normality is 0.991, which also supports our claim.

c-)

```
std <- 3.234 #from Residual standard error in summary(PH.reg)

# Expected frequencies
p25 <- qt(0.25, 14) * std # -2.239
p50 <- qt(0.5, 14) * std # 0
p75 <- qt(0.75, 14) * std # 2.239

# Residuals and expected values table
tbl <- data.frame(residual=ei)
tbl <- tbl %>%
  arrange(residual) %>%
  mutate(rank=seq(1,16)) %>%
  mutate(expected=round(std*qnorm((rank-.375)/(16+.25)),3))

obs25 <- tbl %>% filter(residual <= p25) %>% count() # 4
obs50 <- tbl %>% filter(residual <= p50) %>% count() # 7
obs75 <- tbl %>% filter(residual <= p75) %>% count() # 11

kable(tbl, caption="Residuals and Expected Values under Normality", "latex") %>%
  kable_styling(position="center", latex_options="hold_position")
```

Table 1: Residuals and Expected Values under Normality

	residual	rank	expected
3	-5.150	1	-5.720
16	-3.975	2	-4.145
12	-3.700	3	-3.196
7	-2.425	4	-2.464
1	-2.150	5	-1.841
14	-1.975	6	-1.280
4	-1.150	7	-0.755
13	0.025	8	-0.250
10	0.300	9	0.250
5	0.575	10	0.755
11	1.300	11	1.280
6	2.575	12	1.841
15	3.025	13	2.464
9	3.300	14	3.196
2	3.850	15	4.145
8	5.575	16	5.720

Percentiles	t-value	Expected Freq.	Observed Freq.
25th	-2.239	4	4
50th	0	8	7
75th	2.239	12	11

Since n = 16, the 25th, 50th, and 75th percentiles of the expected frequencies under normality are 4, 8, and 12. The t-value for the corresponding percentiles are -2.239, 0, and 2.239, and the table above shows that the observed frequencies are 4, 7, and 11. The observed and expected frequencies are very close to each other, and it is consistent with part (b) that the normality assumption holds. Also, the residuals and their expected values under normality are close to each other.

d-)

```
# Brown-Forsythe Test
sep <- data.frame(cbind(PH.data, ei))
p1 <- sep[sep[,2] <= 24,]
p2 <- sep[sep[,2] > 24,]
M1 <- summary(p1[,3])[3]
M2 <- summary(p2[,3])[3]
N1 <- dim(p1)[1]
N2 <- dim(p2)[1]

d1 <- abs(p1[,3]-M1)
d2 <- abs(p2[,3]-M2)
s2 <- sqrt((var(d1)*(N1-1)+var(d2)*(N2-1))/(N1+N2-2))

Den <- s2*sqrt(1/N1+1/N2)
Num <- mean(d1)-mean(d2)

T <- Num/Den # 0.856
T.crit <- qt(0.995, N1+N2-2) # 2.977
pvalue <- 2*(1-pt(T, N1+N2-2)) # 0.407
```

H_0: The error variances are constant. H_a: The error variances are not constant.

Since $t^*_{BF} = 0.856 \leq 2.977$ ($\alpha = 0.01$), we conclude H_0, so the error variance is constant and does not vary with the level of X. The p-value of $0.407 \geq \alpha = 0.01$ also supports our claim. This conclusion is consistent with our preliminary findings in part (a).

```
# Using onewaytests library
bf.data<-data.frame(cbind(PH.data,PH.reg$residuals,PH.reg$fitted.values))
dimnames(bf.data)[[2]][3:4]<-c("residuals","fitted.values")
bf.data1<-data.frame(cbind(bf.data,ind=as.factor(I(bf.data$X<=median(bf.data$X))*1)))
bf.test(residuals~ind,data=bf.data1)
```

```
##
##   Brown-Forsythe Test (alpha = 0.05)
## -------------------------------------------------------------
##   data : residuals and ind
##
##   statistic  : 0.06942228
##   num df     : 1
##   denom df   : 13.1945
##   p.value    : 0.7962498
##
##   Result     : Difference is not statistically significant.
## -------------------------------------------------------------
```

The Brown-Forsythe test using 'onewaytests' library gives the same result that we conclude H_0 although the p-value is different from the one above calculated manually as 'onewaytests' uses F-test, rather than t-test.

e-)

```
# Breusch-Pagan Test
ei2 <- ei^2
g <- lm(ei2 ~ PH.data$X)
anova(g)
```

```
## Analysis of Variance Table
##
## Response: ei2
##           Df  Sum Sq Mean Sq F value Pr(>F)
## PH.data$X  1   66.38  66.385  0.7483 0.4016
## Residuals 14 1242.01  88.715
```

```
anova(PH.reg)
```

```
## Analysis of Variance Table
##
## Response: Y
##           Df Sum Sq Mean Sq F value    Pr(>F)
## X          1 5297.5  5297.5  506.51 2.159e-12 ***
## Residuals 14  146.4    10.5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
SSR.S <- anova(g)$`Sum Sq`[1] #66.385
SSE <- anova(PH.reg)$`Sum Sq`[2] #146.425
ChiSquare <- (SSR.S/2) / ((SSE/nrow(PH.data))^2) #0.396
Chi.crit <- qchisq(0.99,1) #6.635
pvalue2 <- 1- pchisq(ChiSquare,1) #0.529
```

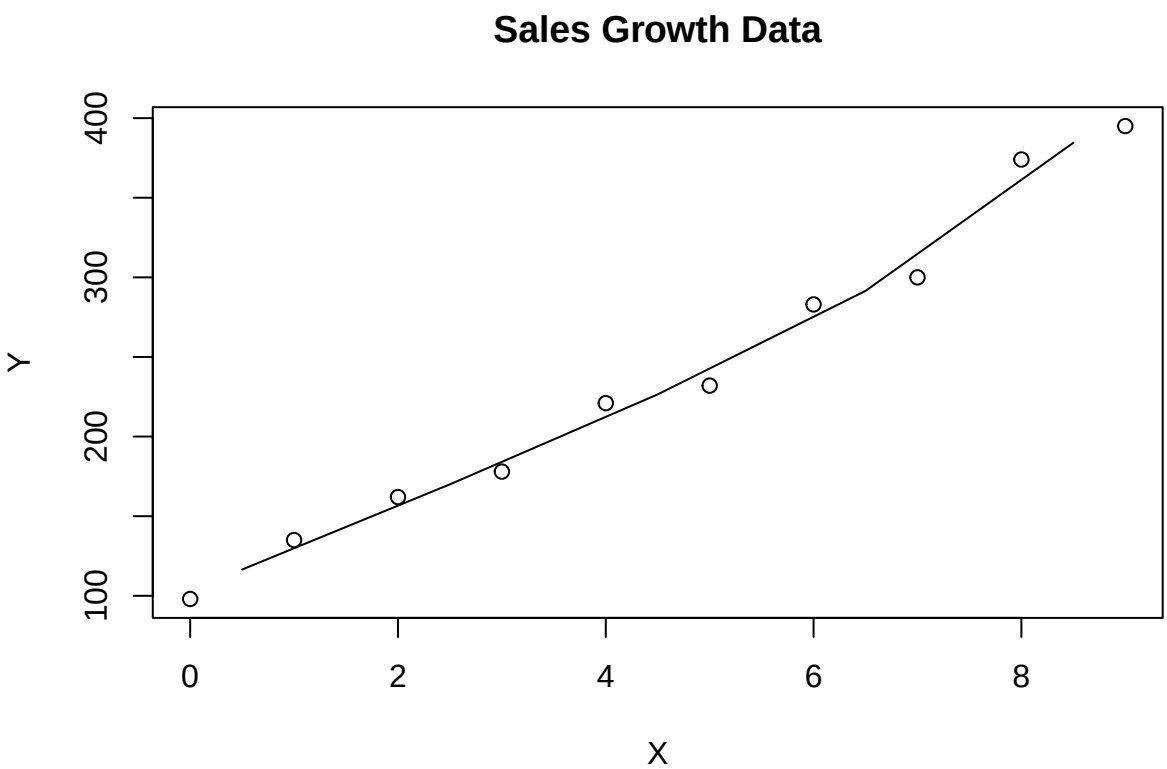
$H_0 : \gamma_1 = 0 \; H_a : \gamma_1 \neq 0$

Since $\chi^2_{BP} = 0.396 \leq 2.977 = \chi^2(0.99, 1)$, we conclude H_0, so the error variance is constant. The p-value of $0.529 \geq \alpha = 0.01$ also supports our claim. This conclusion is consistent with our preliminary findings in part (a).

Problem 2

a-)

```
SG.data <- read.csv("/cloud/project/Data/Sales Growth Data.csv")

band1 <- subset(SG.data, -0.5<X & X<1.5)
band2 <- subset(SG.data, 1.5<X & X<3.5)
band3 <- subset(SG.data, 3.5<X & X<5.5)
band4 <- subset(SG.data, 5.5<X & X<7.5)
band5 <- subset(SG.data, 7.5<X & X<9.5)

median.X <- c(median(band1$X),median(band2$X),median(band3$X),median(band4$X),median(band5$X))
median.Y <- c(median(band1$Y),median(band2$Y),median(band3$Y),median(band4$Y),median(band5$Y))

plot(SG.data$X, SG.data$Y, xlab="X", ylab="Y", main="Sales Growth Data")
lines(median.X, median.Y)
```

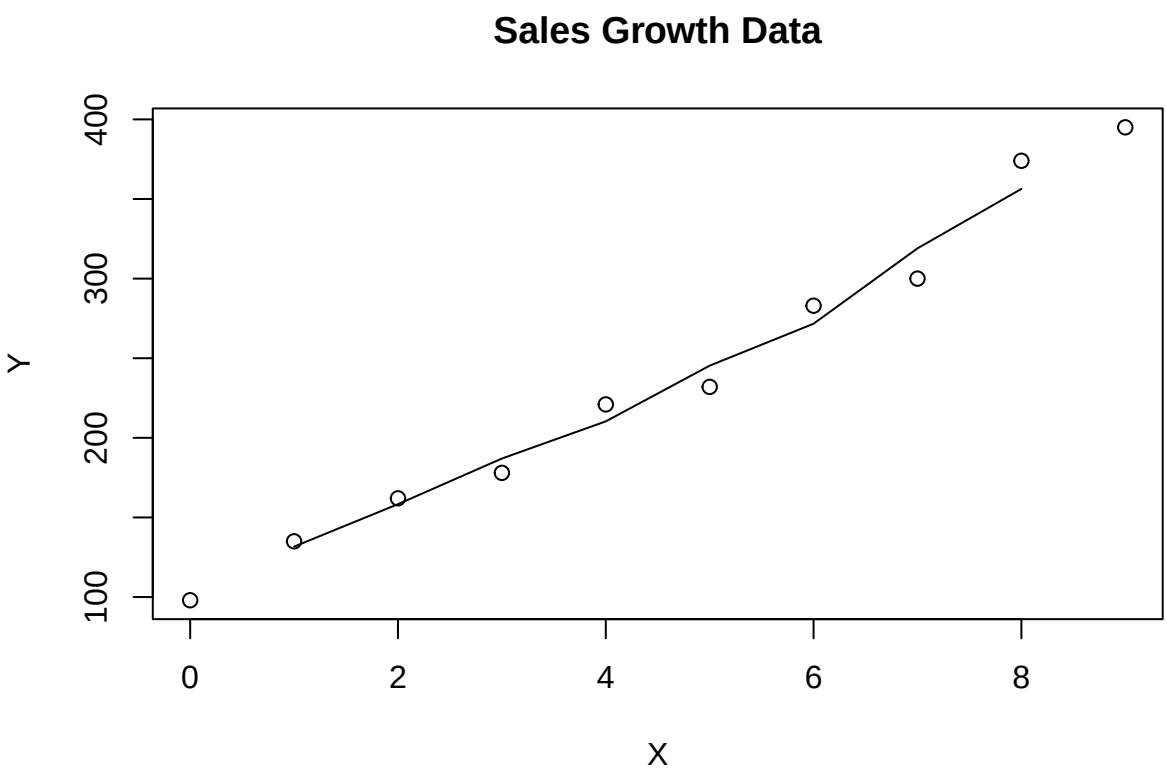


Although there is a slight kink at around X=6.5, the band smooth suggests that the regression relation is mostly linear.

b-)

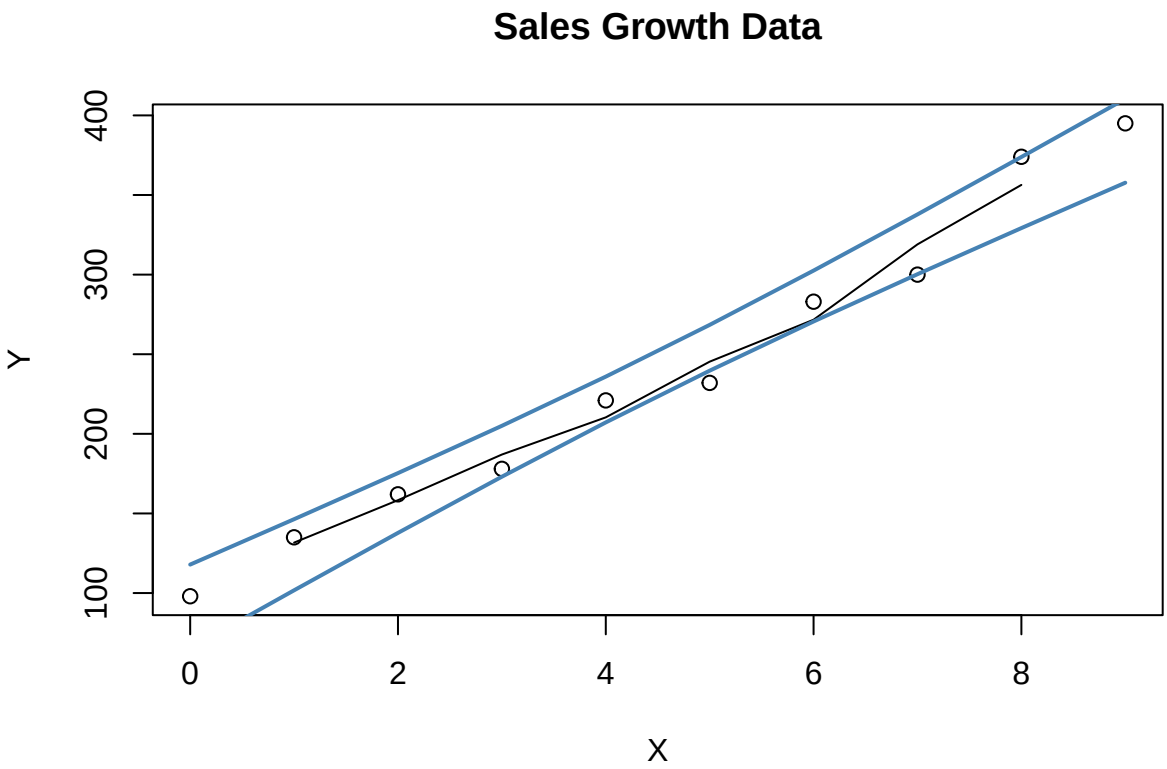
```
band1.1 <- subset(SG.data, -0.5<X & X<2.5)
band2.1 <- subset(SG.data, 0.5<X & X<3.5)
band3.1 <- subset(SG.data, 1.5<X & X<4.5)
```

```r
band4.1 <- subset(SG.data, 2.5<X & X<5.5)
band5.1 <- subset(SG.data, 3.5<X & X<6.5)
band6.1 <- subset(SG.data, 4.5<X & X<7.5)
band7.1 <- subset(SG.data, 5.5<X & X<8.5)
band8.1 <- subset(SG.data, 6.5<X & X<9.5)

fit1 <- lm(Y ~ X, data=band1.1)
fit2 <- lm(Y ~ X, data=band2.1)
fit3 <- lm(Y ~ X, data=band3.1)
fit4 <- lm(Y ~ X, data=band4.1)
fit5 <- lm(Y ~ X, data=band5.1)
fit6 <- lm(Y ~ X, data=band6.1)
fit7 <- lm(Y ~ X, data=band7.1)
fit8 <- lm(Y ~ X, data=band8.1)

M1X <- median(band1.1$X)
M2X <- median(band2.1$X)
M3X <- median(band3.1$X)
M4X <- median(band4.1$X)
M5X <- median(band5.1$X)
M6X <- median(band6.1$X)
M7X <- median(band7.1$X)
M8X <- median(band8.1$X)

Y1 <- predict(fit1,data.frame(X=M1X))
Y2 <- predict(fit2,data.frame(X=M2X))
Y3 <- predict(fit3,data.frame(X=M3X))
Y4 <- predict(fit4,data.frame(X=M4X))
Y5 <- predict(fit5,data.frame(X=M5X))
Y6 <- predict(fit6,data.frame(X=M6X))
Y7 <- predict(fit7,data.frame(X=M7X))
Y8 <- predict(fit8,data.frame(X=M8X))

Xc <- c(M1X,M2X,M3X,M4X,M5X,M6X,M7X,M8X)
Yc_hat <- c(Y1,Y2,Y3,Y4,Y5,Y6,Y7,Y8)

plot(SG.data$X, SG.data$Y, xlab="X", ylab="Y", main="Sales Growth Data")
lines(Xc, Yc_hat)
```



c-)

```
true.reg <- lm(Y ~ X, data=SG.data)
bd<-predict(true.reg,SG.data,se.fit=TRUE)

W=sqrt(2*qf(0.95,2,bd$df))

CB_lower<-bd$fit-W*bd$se.fit
CB_upper<-bd$fit+W*bd$se.fit
dat<-data.frame(X=SG.data$X,CB_lower,CB_upper)

plot(SG.data$X, SG.data$Y, xlab="X", ylab="Y", main="Sales Growth Data")
lines(Xc, Yc_hat)
lines(dat$X,dat$CB_lower,lty=1,lwd=2, col="steelblue")
lines(dat$X,dat$CB_upper,lty=1,lwd=2, col="steelblue")
```

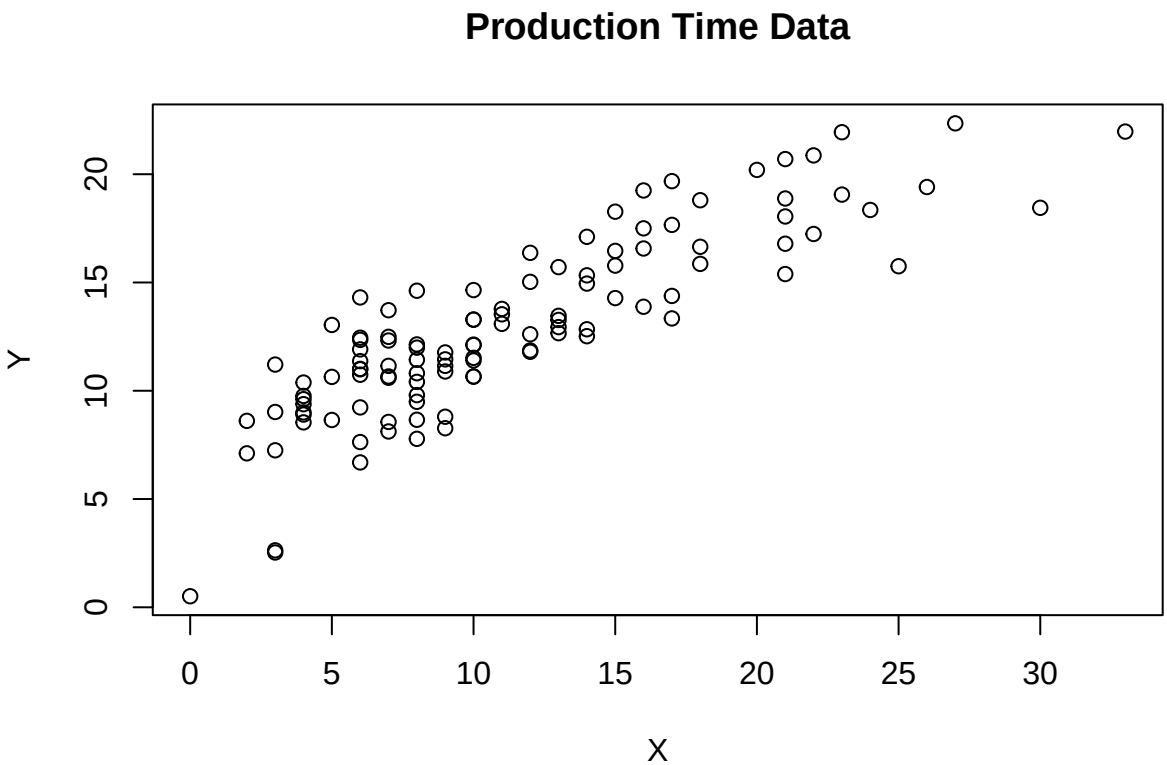


The simplified lowess smooth falls entirely within the confidence band for the regression line, and it shows that the linear regression is appropriate.

Problem 3

a-)

```
PT.data <- read.csv("/cloud/project/Data/Production Time.csv")
plot(PT.data$X, PT.data$Y, xlab="X", ylab="Y", main="Production Time Data")
```



A linear relation does not seem adequate here since the scatter plot above shows a curvilinear shape. The transformation on X would be more appropriate in case of nonlinear relation, which is the case here. In particular, 'log' or 'square root' transformation seems appropriate, given the pattern of the scatter plot.

b-)

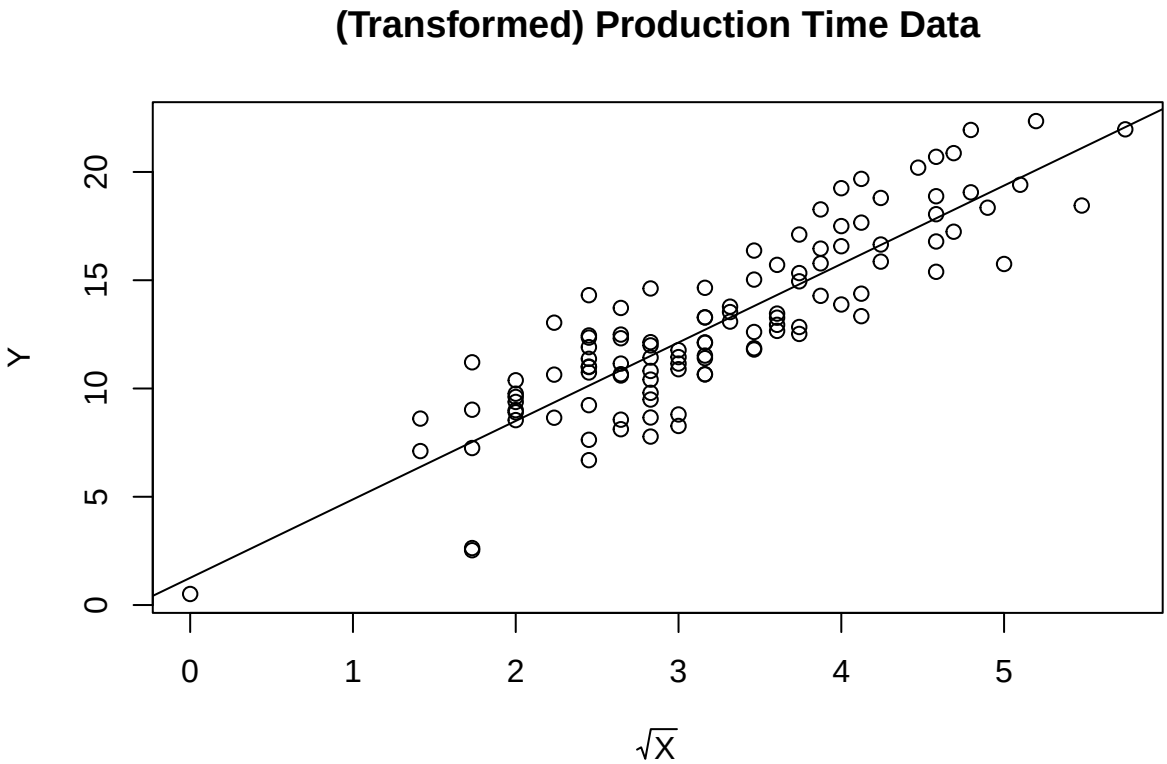
```
trans.X <- sqrt(PT.data$X)
trans.PT.data <- data.frame(cbind(PT.data, trans.X))
trans.PT.reg <- lm(Y ~ trans.X, data=trans.PT.data)
b0.p3 <- trans.PT.reg$coefficients[1]
b1.p3 <- trans.PT.reg$coefficients[2]
summary(trans.PT.reg)
```

```
##
## Call:
## lm(formula = Y ~ trans.X, data = trans.PT.data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.0008 -1.2161  0.0383  1.3367  4.1795
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.2547     0.6389   1.964   0.0521 .
## trans.X       3.6235     0.1895  19.124   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.99 on 109 degrees of freedom
## Multiple R-squared:  0.7704, Adjusted R-squared:  0.7683
## F-statistic: 365.7 on 1 and 109 DF,  p-value: < 2.2e-16
```

The estimated linear regression function for the transformed data:
$X' = \sqrt{X}$
Y = 1.255 + 3.624X'

c-)

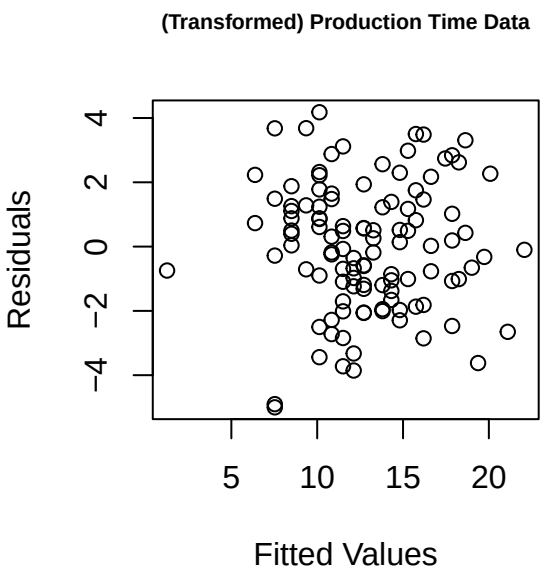
```
plot(trans.PT.data$trans.X, trans.PT.data$Y, xlab=expression(sqrt(X)), ylab="Y",
     main="(Transformed) Production Time Data")
abline(trans.PT.reg)
```

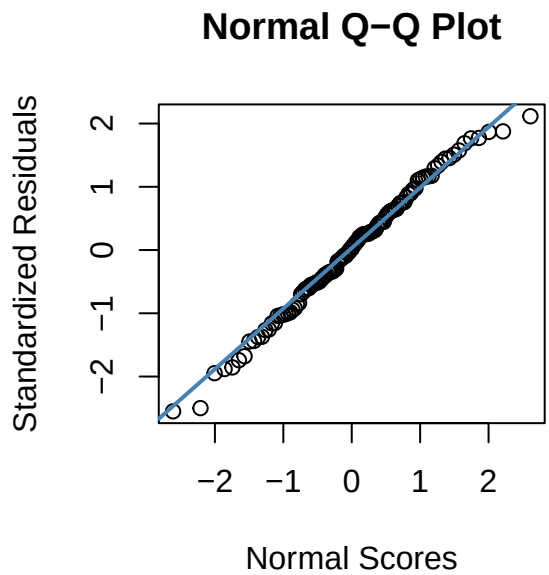


The regression line appear to be a good fit to the transformed data as the plot above shows.

d-)

```
par(mfrow=c(1,2))
ei.p3 <- trans.PT.reg$residuals
fitted.p3 <- trans.PT.reg$fitted.values
plot(fitted.p3, ei.p3, xlab="Fitted Values", ylab="Residuals", main="(Transformed) Production Time Data

PTerror.std <- rstandard(trans.PT.reg)
qqnorm(PTerror.std, xlab="Normal Scores", ylab="Standardized Residuals")
qqline(PTerror.std, col="steelblue", lwd=2)
```





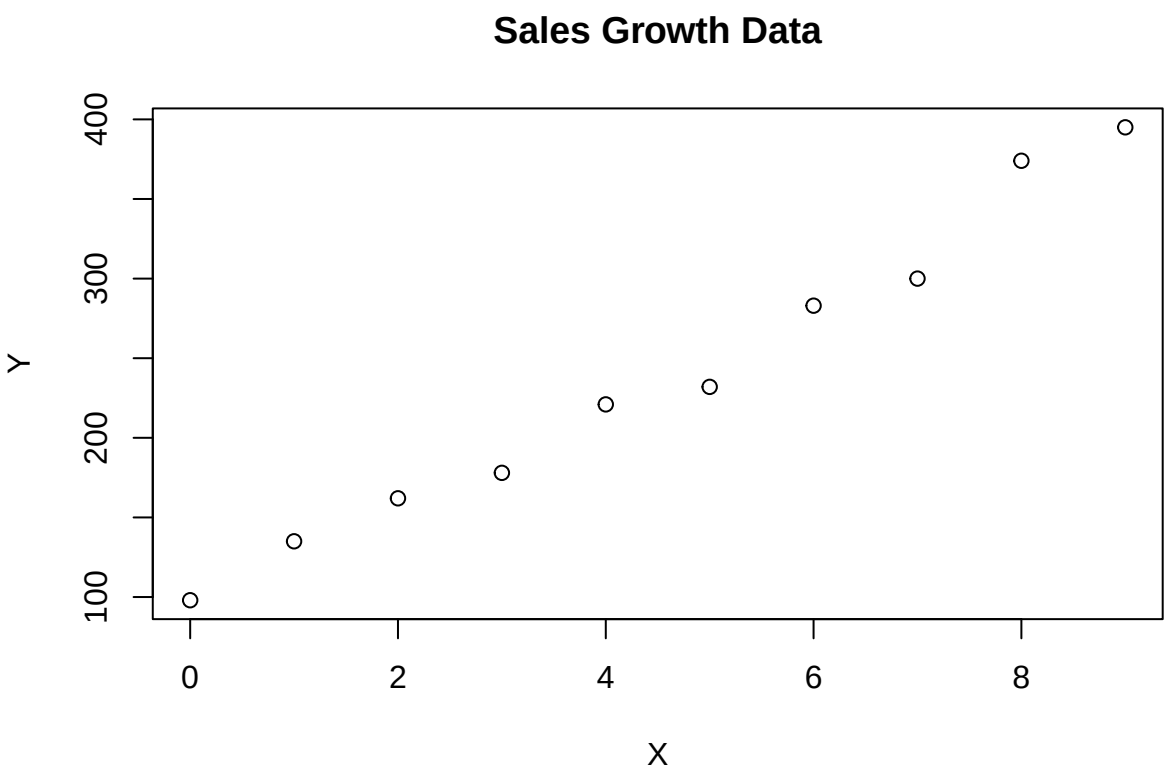
The residuals vs. fitted values plot on the left shows no noticeable pattern, so the error terms seem to have constant variances. The normal probability plot on the right shows that it is approximately normally distributed.

e-) Y = 1.255 + 3.624$\sqrt{X}$

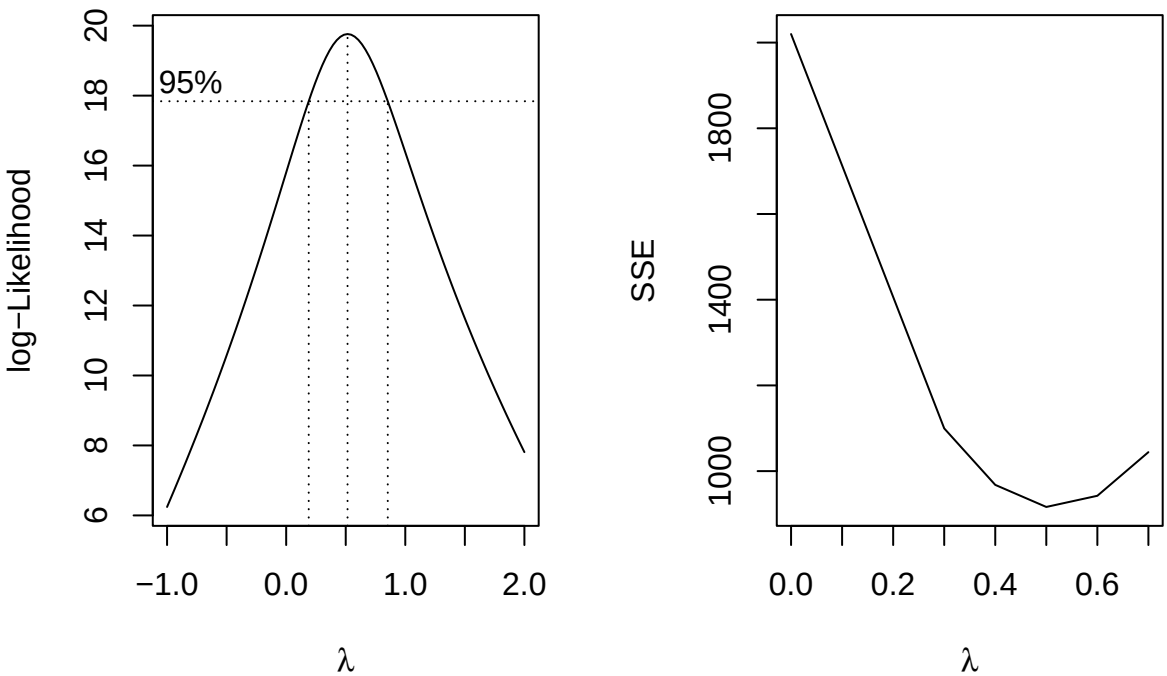
Problem 4

a-)

```
# SG.data and true.reg already exist from P2
plot(SG.data$X, SG.data$Y, xlab="X", ylab="Y", main="Sales Growth Data")
```



```r
par(mfrow=c(1,2))
boxcox(true.reg, lambda=seq(-1,2,by=.1))

# Evaluate SSE
boxcox.sse(SG.data$X, SG.data$Y, l=seq(.3,.7,by=.1))
```




```
##   lambda       SSE
## 6    0.0 2019.8767
## 1    0.3 1099.7093
## 2    0.4  967.9088
## 3    0.5  916.4048
## 4    0.6  942.4498
## 5    0.7 1044.2384
```

The scatter plot shows that a linear relation appears adequate, but the Box-Cox plot indicates $\lambda = 0.5$, which means the square root transformation of Y is suggested. The table above also shows that SSE is minimized when $\lambda = 0.5$. We can infer from this fact needing the transformation of Y that either the error terms do not have constant variances, or it is not normally distributed.

b-)

```r
trans.Y <- sqrt(SG.data$Y)
trans.SG.data <- data.frame(SG.data, trans.Y)
trans.SG.reg <- lm(trans.Y ~ X, data=trans.SG.data)
b0.p4 <- trans.SG.reg$coefficients[1] #10.261
b1.p4 <- trans.SG.reg$coefficients[2] #1.076
summary(trans.SG.reg)
```

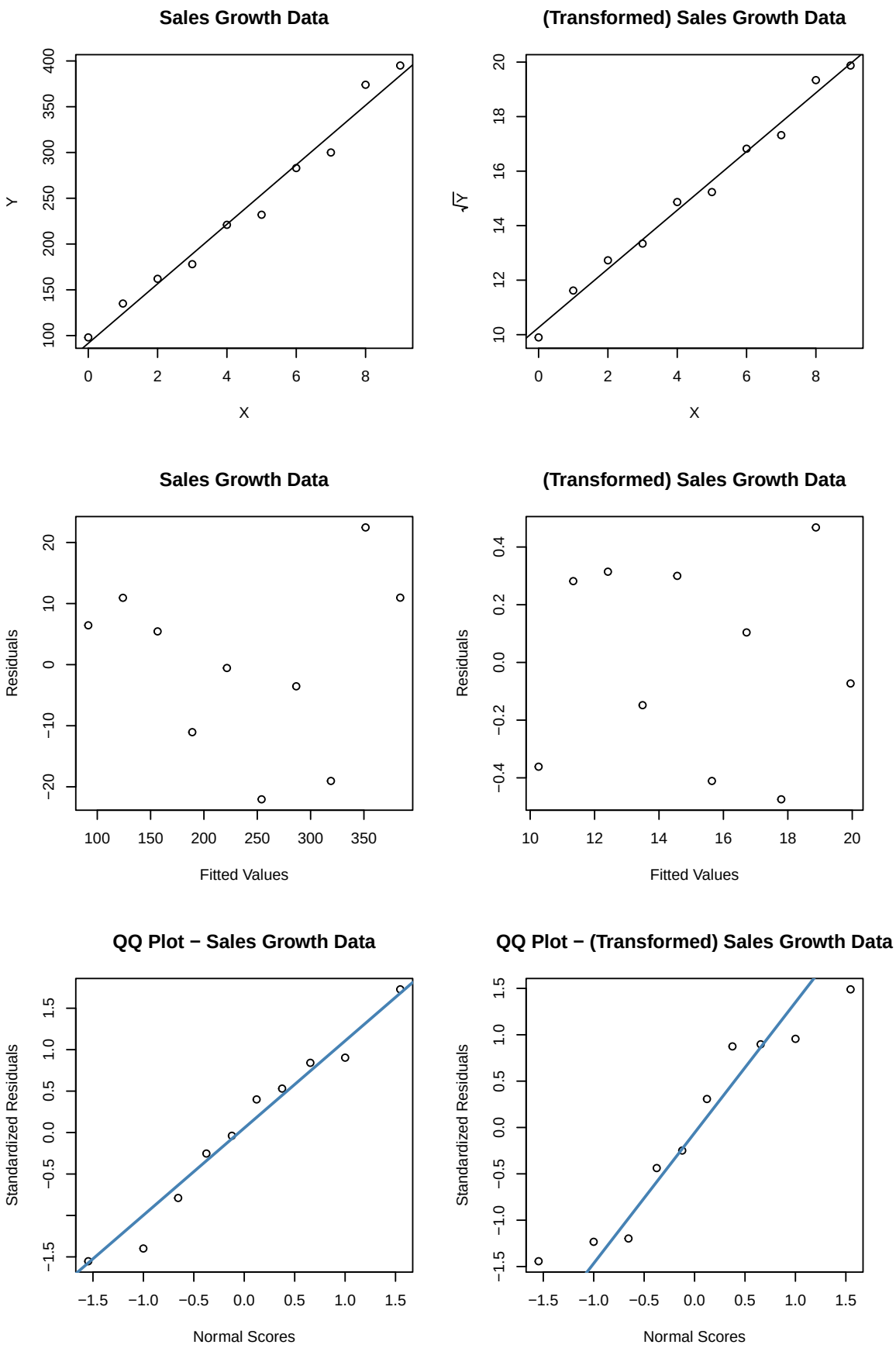
```
##
## Call:
## lm(formula = trans.Y ~ X, data = trans.SG.data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -0.47447 -0.30811  0.01549  0.29541  0.46781
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 10.26093    0.21290   48.20 3.80e-11 ***
## X            1.07629    0.03988   26.99 3.83e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3622 on 8 degrees of freedom
## Multiple R-squared:  0.9891, Adjusted R-squared:  0.9878
## F-statistic: 728.4 on 1 and 8 DF,  p-value: 3.826e-09
```

The estimated linear regression function for the transformed data:
$Y' = \sqrt{Y}$
$Y' = 10.261 + 1.076\text{X}$

c-)

```r
par(mfrow=c(3,2))
# Regression lines
plot(SG.data$X, SG.data$Y, xlab="X", ylab="Y", main="Sales Growth Data")
abline(true.reg)

plot(trans.SG.data$X, trans.SG.data$trans.Y, xlab="X", ylab=expression(sqrt(Y)),
     main="(Transformed) Sales Growth Data")
abline(trans.SG.reg)

# Residuals vs. fitted values
ei.p41 <- true.reg$residuals
fitted.p41 <- true.reg$fitted.values
plot(fitted.p41, ei.p41, xlab="Fitted Values", ylab="Residuals", main="Sales Growth Data")

ei.p42 <- trans.SG.reg$residuals
fitted.p42 <- trans.SG.reg$fitted.values
plot(fitted.p42, ei.p42, xlab="Fitted Values", ylab="Residuals",
     main="(Transformed) Sales Growth Data")

# QQ plot
SGerror.std1 <- rstandard(true.reg)
qqnorm(SGerror.std1, xlab="Normal Scores", ylab="Standardized Residuals",
       main="QQ Plot - Sales Growth Data")
qqline(SGerror.std1, col="steelblue", lwd=2)

SGerror.std2 <- rstandard(trans.SG.reg)
qqnorm(SGerror.std2, xlab="Normal Scores", ylab="Standardized Residuals",
       main="QQ Plot - (Transformed) Sales Growth Data")
qqline(SGerror.std2, col="steelblue", lwd=2)
```


Sales Growth Data
Y
X
(Transformed) Sales Growth Data
$\sqrt{Y}$
X
Sales Growth Data
Residuals
Fitted Values
(Transformed) Sales Growth Data
Residuals
Fitted Values
QQ Plot − Sales Growth Data
Standardized Residuals
Normal Scores
QQ Plot − (Transformed) Sales Growth Data
Standardized Residuals
Normal Scores

The scatter plots on the top indicate that both the original and transformed data appear to be a good fit to the corresponding linear regression although the transformed model has a better fit.

The residuals vs. fitted values plots in the middle indicate that the variances of error terms are not constant in the original data whereas those seem constant in the transformed data.

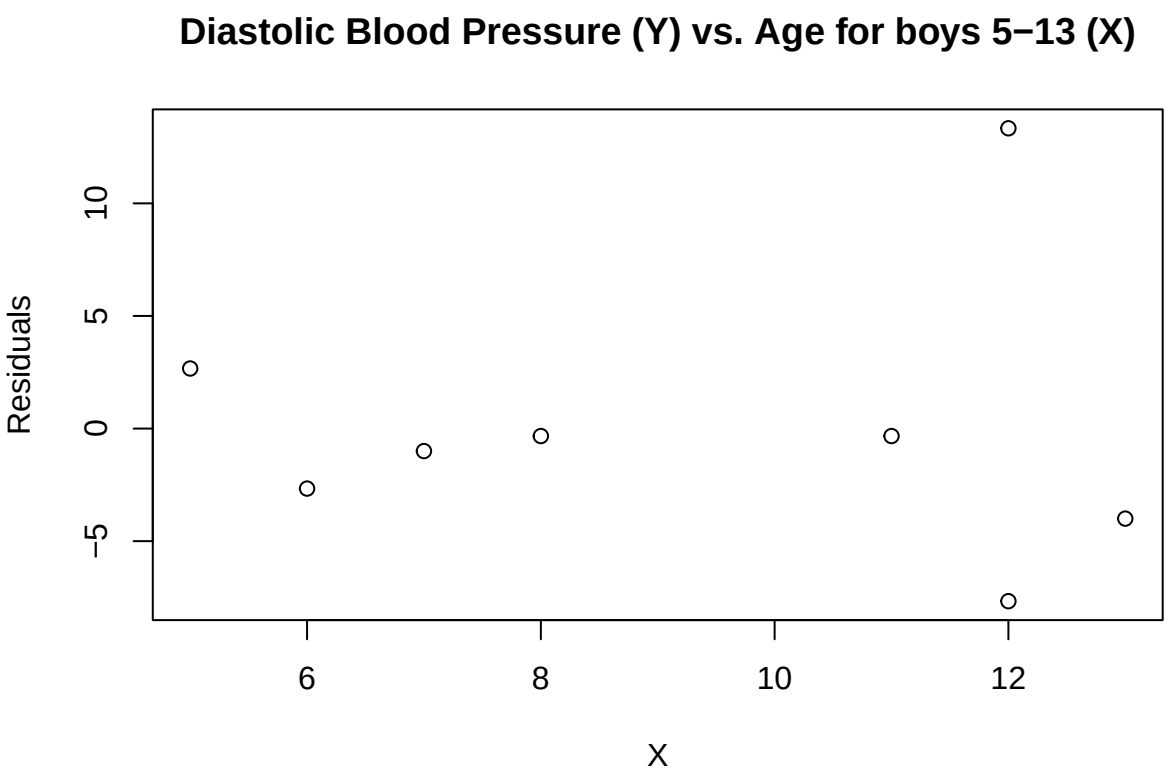
The normal probability plots on the bottom indicate that both the original and transformed data are approximately normally distributed, but we can notice a slightly curvy shape in the plot of the original data.

d-) $\sqrt{Y} = 10.261 + 1.076X$

Problem 5

a-)

```
X <- c(5,8,11,7,13,12,12,6) #diastolic blood pressure
Y <- c(63,67,74,64,75,69,90,60) #age for boys 5 to 13
p5.data <- data.frame(cbind(X, Y))
p5.reg <- lm(Y ~ X, data=p5.data)
b0.p5 <- p5.reg$coefficients[1] #48.667
b1.p5 <- p5.reg$coefficients[2] #2.333

ei.p5 <- p5.reg$residuals
plot(p5.data$X, ei.p5, xlab="X", ylab="Residuals", main="Diastolic Blood Pressure (Y) vs. Age for boys 5
```



The residual plot above shows that the error terms do not have constant variances and seem to have larger variances where X is bigger.

The estimated regression function: Y = 48.667 + 2.333X

b-)

```
p5.newdata <- p5.data[-7,]
p5.newreg <- lm(Y ~ X, data=p5.newdata)
b0.p5new <- p5.newreg$coefficients[1] #53.068
b1.p5new <- p5.newreg$coefficients[2] #1.621
```

The estimated regression function: Y = 53.068 + 1.621X

After omitting the case 7 from the data, the regression function has smaller b_1, so the case 7 affected the slope of the initial regression function to be steeper. We could have expected it by seeing the original data because Y=90 at case 7 whereas Y sits 60 ~ 75 for all other cases. The case 7 seems to be an outlier.

c-)

```
predict.int <- predict(p5.newreg, data.frame(X=12), interval="prediction", level=0.99)
lwr <- predict.int[2] #60.313
upr <- predict.int[3] #84.736
```

99% Prediction interval for a new Y observation at X=12: $60.313 \leq Y_{h(new)} \leq 84.736$

The observation Y_6 falls inside this prediction interval. Falling inside the prediction interval is not very meaningful as it is designed to be rather wide and to contain most of the observations at X=12.

If the observation actually fell outside the prediction interval as the observation Y_7 does, we should be alerted that maybe the blood pressure was measured incorrectly, or that 12-year boy has something going on in his body, etc.